An Unsupervised Deep Unfolded Network for Mosaic-Based Hyperspectral Fusion Imaging

Nan Wang, Aiping Mo, Renwei Dian, *Member, IEEE*, Jujue Wei, and Shutao Li, *Fellow, IEEE*

***Abstract*—Snapshot hyperspectral imaging based on a mosaic sensor enables video-rate acquisition but suffers from inherently low spatial resolution. A practical solution is to fuse the low-resolution (LR) mosaiced hyperspectral image (HSI) with a high-resolution (HR) panchromatic (PAN) image of the same scene. However, recovering an HR HSI from such observations is a severely illposed inverse problem, especially in the absence of ground-truth supervision. In this article, we propose an unsupervised deep unfolded network (UDUN) for the mosaiced and PAN image fusion. To regularize the immense solution space of this illposed inverse problem, we exploit the synergy between the deep unfolded network (DUN) and the transformation equivariance prior (TEP), where the explicit degradation modeling capability of DUN is leveraged to deliver an accurate estimation of the spectral response function (SRF), thereby enabling the effective incorporation of TEP as a geometrically consistent and physically grounded prior in unsupervised training. Extensive experiments on three widely used HSI datasets demonstrate that the proposed UDUN achieves superior reconstruction quality and robustness compared with state-of-the-art fusion methods, highlighting its effectiveness and generalization capability.**

***Index Terms*—Deep unfolded network (DUN), hyperspectral imaging, mosaiced and panchromatic (PAN) image fusion, transformation equivariance prior (TEP).**

I. Introduction

HYPERSPECTRAL imaging has emerged as a pivotal technology in remote sensing, distinguished by its ability to capture rich spectral signatures across dozens of continuous bands [1], [2], [3]. This capability enables fine-grained material identification and scene analysis, finding broad applications in agriculture [4], [5], environmental monitoring [6], [7], [8], [9], and mineral exploration [10], [11], [12]. Traditionally, hyperspectral images (HSIs) are acquired using push-broom or whisk-broom scanners. While these line-scanning devices offer high spatial and spectral resolution, they require a static scene or a stable platform to accumulate data over time. Consequently, this sequential scanning mechanism results in low temporal resolution and susceptibility to motion artifacts, rendering them unsuitable for dynamic scene observation or time-sensitive applications.

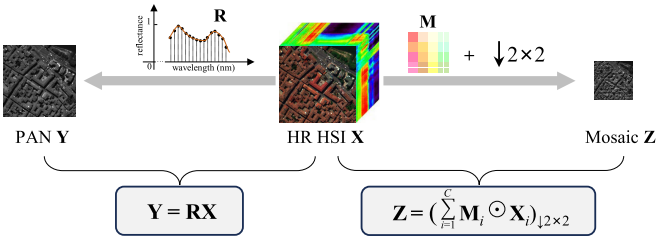


Fig. 1. Illustration of the hyperspectral imaging via the mosaiced and PAN image fusion, along with the degradation relationship between the ideal HR HSI and the observed mosaiced and PAN images. Note that the PAN image exhibits twice the height and width of the mosaiced image. Both the mosaiced and PAN images exhibit only one channel at acquisition.

To overcome the limitations of scanning-based systems, snapshot compressive imaging (SCI) was developed to capture 3-D hyperspectral data in a single shot [13], [14], [15], [16]. A prominent implementation involves mosaiced image sensors utilizing a spectral filter array (SFA), enabling video-rate HSI acquisition [17], [18]. However, governed by the inherent tradeoff between spatial and spectral resolution on a 2-D sensor, the resulting mosaiced image inevitably suffers from low spatial resolution. A practical solution to mitigate this is fusing the low-resolution (LR) mosaiced HSI with a high-resolution (HR) panchromatic (PAN) image of the same scene. As illustrated in Fig. 1, the reconstructed HR HSI exhibits both high spatial and spectral fidelity. Despite the effectiveness of this strategy, recovering HR HSI from raw observations remains challenging, since it is a severely illposed inverse problem.

Existing approaches to this fusion problem can be broadly categorized into traditional methods and deep learning (DL)-based methods. Traditional methods typically rely on hand-crafted priors [19], such as matrix factorization [20], tensor decomposition [21], and sparse coding [22]. Although

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The code is available at https://github.com/Nan-Wong98/An-Unsupervised-Deep-Unfolded-Network-for-Mosaic-based-Hyperspectral-Fusion-Imaging

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mathematically grounded, these methods often lack sufficient representation capability to model complex spectral–spatial correlations, limiting their reconstruction quality. Conversely, DL-based methods [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35] have achieved superior performance by learning mappings from paired training samples. These can be further divided into supervised and unsupervised categories. Supervised methods require paired ground-truth HR HSI for training [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47]. However, acquiring HR HSI ground truth in real-world scenarios is notoriously difficult and expensive. Consequently, unsupervised methods [48], [49], [50], [51], which rely solely on observations without reference, are more practical and generalizable for real-world applications.

Recently, the transformation equivariance prior (TEP) has proven highly effective in various image restoration tasks [49], [52], [53], especially mosaiced and PAN image fusion [54], [55]. Its core principle lies in enforcing a consistent geometric correspondence between the network input and output. For instance, if the input undergoes a specific rotation, the reconstruction should exhibit a corresponding rotation. Despite its efficacy, the successful application of TEP is strictly contingent upon the accurate estimation of the degradation function. In particular, if the degradation relationship is inaccurately estimated, the equivariance constraint may be applied to an incorrect target, leading to optimization divergence or structural distortions. This limitation highlights that the full potential of TEP can be unleashed when the degradation function is estimated accurately.

Deep unfolded networks (DUNs) have garnered significant attention [56], [57], [58], [59], [60], [61], [62], [63], [64] for their ability to unfold iterative optimization algorithms into neural network layers. This architecture offers transparent internal mechanisms and, crucially, an explicit representation of the physical degradation process. By embedding the degradation operator directly into the network, DUNs possess the intrinsic capability to learn and refine degradation parameters during restoration. This characteristic creates a natural and powerful synergy with TEP. In particular, the DUN framework ensures the accurate estimation of the degradation function, which is the prerequisite for TEP. In return, TEP provides a robust geometric prior to constrain the solution space of the DUN, effectively mitigating the illposed nature of unsupervised learning. This complementary relationship suggests that integrating TEP into a degradation-aware DUN framework is a promising direction for achieving superior reconstruction performance.

In this article, we propose an unsupervised deep unfolded network (UDUN) for the fusion of mosaiced and PAN images. By leveraging the inherent synergy within the architecture, where the DUN explicitly models the physical degradation, we overcome the barrier to applying the TEP in unsupervised settings with unknown degradation. This integration enforces strong geometric consistency alongside observation fidelity, significantly mitigating artifacts and improving convergence stability without the need for ground-truth data. The main contributions of this article are summarized as follows.

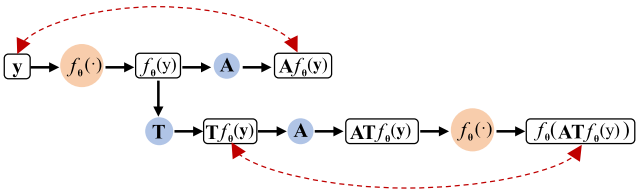


Fig. 2. Illustration of the TEP-based training framework. The dashed red lines represent calculating the loss function. Apart from observation consistency, the transformed reconstruction is degraded into pseudo-observations and then rereconstructed by the fusion network. Enforcing consistency between the transformed reconstruction and the rereconstructed result would effectively strengthen the reconstruction capacity of the network.

1) We reveal a synergistic mechanism where the DUN estimates unknown degradation to enable TEP application, while TEP conversely imposes geometric constraints to regularize the unsupervised learning and mitigate artifacts.
2) We propose the UDUN for the mosaiced and PAN image fusion, which jointly estimates the HR HSI and spectral response function (SRF) to achieve high-fidelity reconstruction without ground-truth data.
3) Extensive experiments on three widely used HSI datasets demonstrate the effectiveness and generalization capability of the proposed method.

The remainder of this article is organized as follows. Section II reviews the related works. Section III describes the proposed UDUN. Section IV presents a comprehensive experimental analysis to validate the framework's effectiveness. Section V draws a conclusion of this article.

II. Related Work

A. TEP-Based Fusion Method

The concept of transformation equivariance posits that the output of a function should exhibit a predictable geometric change in response to transformations applied to its input. Mathematically, given a mapping function $f(\cdot)$ and a geometric transformation operator $T(\cdot)$ (e.g., rotation or flipping), equivariance is defined as $f(T(x)) = T(f(x))$. In DL, this principle is widely exploited as a geometric prior to bolster model robustness and mitigate dependence on massive training datasets. As illustrated in Fig. 2, TEP enforces consistency between the transformed reconstruction and the rereconstructed result, thereby providing a geometric constraint for unsupervised image restoration. For example, Wang et al. [54] proposed an unsupervised fusion framework (EFN) that enforced TEP through a consistency loss, minimizing the discrepancy between the transformed reconstruction and the reconstruction of the transformed input, delivering great fusion performance. Furthermore, Dian et al. [55] proposed an equivariant Bayesian variational inference framework (EBVIF) that integrated TEP with learnable physical degradation models, to enable high-fidelity fusion under the noisy condition.

However, there remains substantial room for improving the accuracy of degradation estimation. Since the efficacy of TEP-based frameworks is strictly contingent upon the precision of the estimated degradation, further refining this

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estimation process is critical for unlocking the full potential of equivariant learning.

B. DUNs for HSI Reconstruction

DUNs have garnered significant attention in HSI reconstruction tasks, including HSI superresolution, denoising, and fusion, owing to their ability to bridge the gap between model-based interpretability and data-driven representation learning. By mapping the iterations of traditional optimization algorithms (e.g., ISTA, ADMM) into cascading neural network layers, DUNs inherit the clear physical structure of mathematical models while leveraging the powerful fitting capability of DNNs.

Most existing DUNs for HSI reconstruction operate in a supervised manner, relying on pairs of observations and ground-truth HR HSI to train the network parameters end-to-end [58], [59], [60]. However, the heavy reliance on ground-truth samples restricts their flexibility, as acquiring high-quality reference images is often impractical or expensive. This dependence limits the potential of supervised methods in scenarios where only observed LR data is available. Consequently, the research focus has shifted toward unsupervised DUNs. These methods typically construct loss functions based on observation consistency prior (OCP), i.e., enforcing that the degraded version of the reconstructed HSI matches the raw observations. For instance, recent works have integrated learnable degradation operators within the unfolded framework to jointly estimate the HSI and the degradation function [61], [62], [63]. For example, Qu et al. [61] embedded learnable degradation operators into an unsupervised unfolded framework and optimized the reconstruction by enforcing observation consistency between the estimated HSI and the raw measurements, with several auxiliary regularization terms introduced to stabilize the optimization. Similarly, Li et al. [63], and Cao et al. [64] formulated HSI fusion under a model-informed unsupervised paradigm, where the degradation model serves as the primary supervision signal, while implicit priors induced by network architecture are employed as supplementary regularization.

However, relying primarily on OCP constitutes a weak supervision signal for such a highly illposed inverse problem. Without additional effective regularization, the existing unsupervised DUNs often suffer from artifacts, yielding unpleasant reconstruction performance. In this article, we uncover a synergistic mechanism between DUN and TEP and leverage it to develop the proposed UDUN, demonstrating great fusion performance.

III. Methodology

A. Problem Formulation

To describe the proposed method, we denote the HR HSI as $\mathbf{X} \in \mathbb{R}^{C\times N}$, where C and N denote its spectral and spatial dimensions, respectively. Furthermore, we denote the PAN and mosaiced images as $\mathbf{Y} \in \mathbb{R}^{1\times N}$ and $\mathbf{Z} \in \mathbb{R}^{1\times n}$, where $n = N/4$ is the spatial dimension of the mosaiced image. Since both the mosaiced and PAN images are acquired via single-shot imaging, they possess only a single channel.

The degradation relationship between $\mathbf{Y}$ and $\mathbf{X}$ is defined as

$$\mathbf{Y} = \mathbf{R}\mathbf{X} + \mathbf{N}_{\mathbf{Y}} \tag{1}$$

where $\mathbf{R} \in \mathbb{R}^{1\times C}$ denotes the spectral response vector and $\mathbf{N}_{\mathbf{Y}} \in \mathbb{R}^{1\times N}$ represents noise associated with the PAN image.

Meanwhile, the degradation relationship between $\mathbf{Z}$ and $\mathbf{X}$ is defined as

$$\mathbf{Z} = \left(\sum_{i=1}^{C} \mathbf{M}_i \odot \mathbf{X}_i\right)_{\downarrow 2\times 2} + \mathbf{N}_{\mathbf{Z}} \tag{2}$$

where $\mathbf{M}_i \in \mathbb{R}^N$ and $\mathbf{X}_i \in \mathbb{R}^N$ denote mosaicing pattern and data of the i_{th} band, respectively. $\odot$ is the elementwise multiplication and $(\cdot)_{\downarrow 2\times 2}$ represents the 2×2 average pooling. $\mathbf{N}_{\mathbf{Z}} \in \mathbb{R}^{1\times n}$ denotes noise associated with the mosaiced image. For simplicity, noise terms are omitted in the subsequent derivation.

Given these degradation models, reconstructing the HR HSI $\mathbf{X}$ from the observed $\mathbf{Y}$ and $\mathbf{Z}$ is a severely illposed inverse problem. To address this, we formulate the fusion task as the following optimization problem:

$$\min_{\hat{\mathbf{X}},\hat{\mathbf{R}}} \left\|\mathbf{Y} - \hat{\mathbf{R}}\hat{\mathbf{X}}\right\|_F^2 + \left\|\mathbf{Z} - \left(\sum_{i=1}^{C} \mathbf{M}_i \odot \hat{\mathbf{X}}_i\right)_{\downarrow 2\times 2}\right\|_F^2 + \lambda\Phi\left(\hat{\mathbf{X}}\right) \tag{3}$$

where $\|\cdot\|_F^2$ denotes the square of the Frobenius norm. The first two terms ensure observation consistency, while the third term $\lambda\Phi(\hat{\mathbf{X}})$ serves as a regularization term with weight λ. $\hat{\mathbf{X}}$ and $\hat{\mathbf{R}}$ represent the variable to be estimated since they are inaccessible.

To solve the problem, we propose UDUN for the fusion of mosaiced and PAN images. In particular, we iteratively optimize the predicted HR HSI to minimize reconstruction errors. As illustrated in Fig. 3, within each iteration, the current HSI estimate is degraded to simulate mosaiced and PAN observations, allowing for gradient computation in the observation domain. Subsequently, a DNN is employed to project this gradient from the observation domain back to the target domain, updating the estimate via a residual connection. Through multiple iterations, the reconstruction error is minimized, yielding an HR HSI that closely approximates the target. However, relying primarily on observation consistency terms often hinders convergence. To facilitate unsupervised training and enhance reconstruction performance, we incorporate TEP as the regularization term.

Next, we will detail the workflow of the deep unfolded fusion network.

B. Deep Unfolded Fusion Network

As illustrated in Fig. 3, the deep unfolded fusion network unfolds a traditional iterative optimization process into a deep neural network, where each stage corresponds to one iteration step. Consequently, the output of each iteration refines the previous estimate, allowing the network to progressively converge to the final HR HSI after T iterations. In this article, we set T to two experiments, which is verified in Section IV-F.

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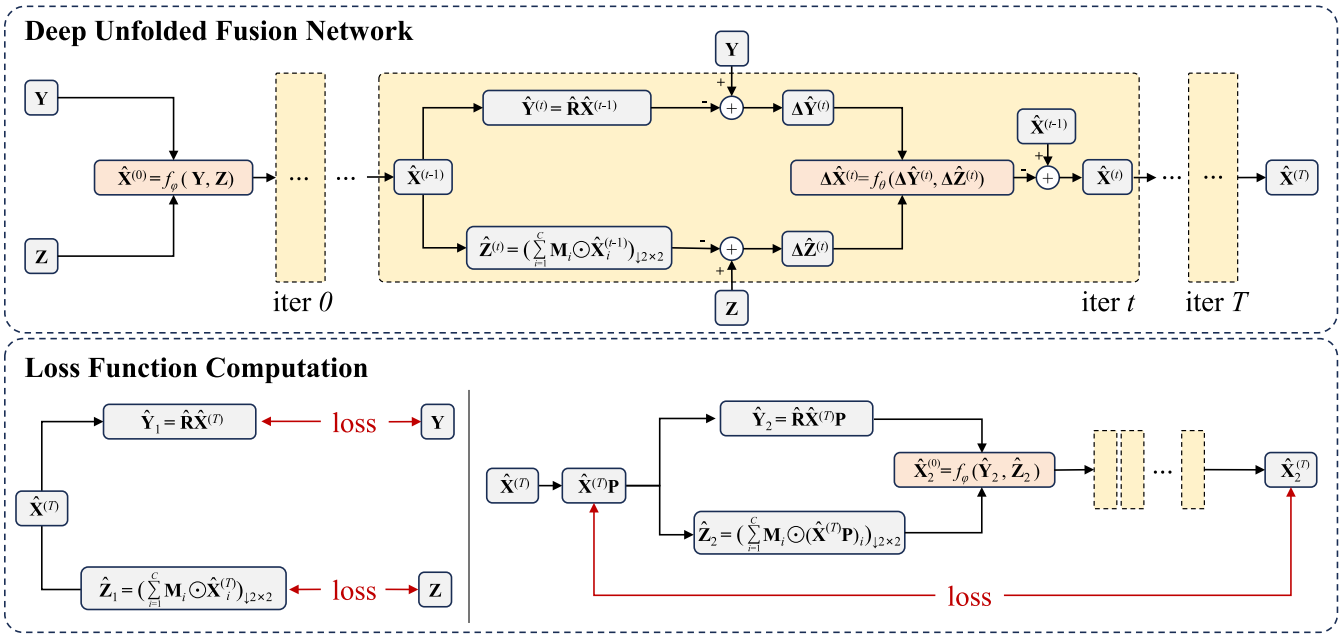


Fig. 3. Schematic of the UDUN for mosaiced and PAN image fusion. (Top) Half presents the forward inference pipeline of the network. (Bottom) Half details the computation flow of the loss function.

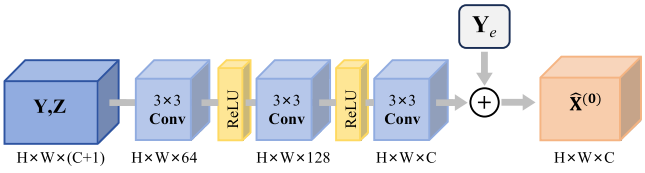


Fig. 4. Structure of shallow fusion network $f_\phi(\cdot)$.

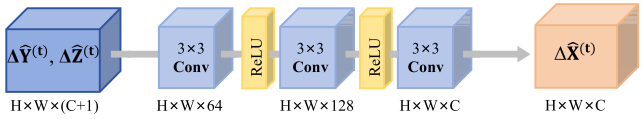


Fig. 5. Structure of gradient projection network $f_\theta(\cdot)$.

First, we derive an initial prediction $\hat{\mathbf{X}}^{(0)} \in \mathbb{R}^{C\times N}$ from the mosaiced image $\mathbf{Z}$ and the PAN image $\mathbf{Y}$ via a shallow fusion network

$$\hat{\mathbf{X}}^{(0)} = f_\phi(\mathbf{Y}, \mathbf{Z}) \tag{4}$$

where $f_\phi(\cdot)$ denotes a DNN parameterized by ϕ. Although $\hat{\mathbf{X}}^{(0)}$ approximates the target HR HSI, it inevitably exhibits reconstruction errors that require further refinement.

The architecture of $f_\phi(\cdot)$ is depicted in Fig. 4. It consists primarily of three sequential convolutional layers, each utilizing a 3×3 kernel, 1×1 stride, and 1×1 padding. The channel depths for these layers are 64, 128, and C, respectively, with ReLU activation applied after the first two convolutional layers, while the last convolutional layer directly produces the output. To prepare the input for this network, we first reshape the single-channel mosaiced image by organizing spectral data into separate channels, generating an LR HSI cube of dimension $\mathbb{R}^{C\times n/16}$. This cube is then interpolated to match the spatial resolution of the PAN image (scale factor of 8×8), resulting in an upsampled mosaiced image of dimension $\mathbb{R}^{C\times N}$. Finally, the concatenation of the PAN image and the upsampled mosaiced image along the channel dimension serves as the input to $f_\phi(\cdot)$. To generate the final output, the PAN image is replicated along the spectral dimension (i.e., $\mathbf{Y}_e \in \mathbb{R}^{C\times N}$) and added to the network's output. Consequently, the network outputs the shallow fused estimate $\hat{\mathbf{X}}^{(0)}$, which serves as the starting point for the iterative optimization.

To describe the process of the iterative optimization, let us take the *tth* iteration as an example. The estimate from the previous step, i.e., $\hat{\mathbf{X}}^{(t-1)}$, is degraded to produce the pseudo-observations $\hat{\mathbf{Y}}^{(t)}$ and $\hat{\mathbf{Z}}^{(t)}$

$$\hat{\mathbf{Y}}^{(t)} = \hat{\mathbf{R}}\hat{\mathbf{X}}^{(t-1)}, \quad \hat{\mathbf{Z}}^{(t)} = \left(\sum_{i=1}^{C} \mathbf{M}_i \odot \hat{\mathbf{X}}_i^{(t-1)}\right)_{\downarrow 2\times 2}. \tag{5}$$

These pseudo-observations are compared with the raw observed data to compute the gradient terms in the observation domain

$$\Delta\hat{\mathbf{Y}}^{(t)} = \mathbf{Y} - \hat{\mathbf{Y}}^{(t)}, \quad \Delta\hat{\mathbf{Z}}^{(t)} = \mathbf{Z} - \hat{\mathbf{Z}}^{(t)}. \tag{6}$$

The gradients are then fed into a gradient projection network, $f_\theta(\cdot)$, to generate the update term $\Delta\hat{\mathbf{X}}^{(t)}$ in the target domain

$$\Delta\hat{\mathbf{X}}^{(t)} = f_\theta\left(\Delta\hat{\mathbf{Y}}^{(t)}, \Delta\hat{\mathbf{Z}}^{(t)}\right) \tag{7}$$

where $f_\theta(\cdot)$ is parameterized by θ.

The structure of $f_\theta(\cdot)$, as shown in Fig. 5, follows the same three-convolution design as $f_\phi(\cdot)$, with ReLU activation applied after the first two convolutional layers and the last convolutional layer directly producing the update term. The main difference is that $f_\theta(\cdot)$ does not employ a skip connection. This is designed because the iterative update rule inherently constitutes a global residual learning process, allowing $f_\theta(\cdot)$ to focus solely on learning the gradient projection mapping

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without redundant internal shortcuts. For reproducibility, both $f_\phi(\cdot)$ and $f_\theta(\cdot)$ adopt three convolutional layers with kernel size 3×3, stride 1, and padding 1. The channel dimensions are $(C+1) \to 64 \to 128 \to C$, where $C = 16$ in our experiments, and ReLU activation is used after the first two convolutional layers.

Finally, the HR HSI estimate is updated as

$$\hat{\mathbf{X}}^{(t)} = \hat{\mathbf{X}}^{(t-1)} + \eta \Delta \hat{\mathbf{X}}^{(t)} \tag{8}$$

where η denotes the learning rate, which controls the step size of the update for the HR HSI estimate in each iteration.

After T iterations, the fusion network could progressively refine the fused image to obtain $\hat{\mathbf{X}}^{(T)}$, which is close to the target HR HSI $\mathbf{X}$. Note that the gradient projection network $f_\theta(\cdot)$ is shared across all the iterations, which ensures consistent gradient projection rules throughout the iterative refinement process and enhances the convergence stability of the fusion model.

Regarding the SRF $\hat{\mathbf{R}}$, it is implemented with a single 3×3 convolutional layer, enforce shared $\hat{\mathbf{R}}$ across all T iterations. Furthermore, this shared $\hat{\mathbf{R}}$ is utilized in the subsequent loss function calculation. The effectiveness of this parameter-sharing strategy is validated in Section IV-G.

C. Loss Functions

To effectively constrain the training of UDUN in unsupervised scenarios, we design a composite loss function that leverages two core priors, i.e., OCP and TEP. The overall framework of these loss functions is illustrated in the bottom panel of Fig. 3.

1) Observation Consistency Prior: In the unsupervised scenario, the absence of ground-truth HR HSI necessitates a training constraint based on observation consistency. This prior relies on the physical assumption that the predicted HR HSI, when subjected to the accurate degradation operators, should faithfully reconstruct the raw observations.

Given the final fused estimate $\hat{\mathbf{X}}^{(T)}$, we simulate the degradation process to generate the corresponding pseudo-observations

$$\hat{\mathbf{Y}}_1 = \hat{\mathbf{R}}\hat{\mathbf{X}}^{(T)}, \quad \hat{\mathbf{Z}}_1 = \left(\sum_{i=1}^{C} \mathbf{M}_i \odot \hat{\mathbf{X}}_i^{(T)} \right)_{\downarrow 2\times 2}. \tag{9}$$

Based on these, the OCP-based loss is formulated as

$$\mathcal{L}_{OC} = \left\| \mathbf{Y} - \hat{\mathbf{Y}}_1 \right\|_F^2 + \left\| \mathbf{Z} - \hat{\mathbf{Z}}_1 \right\|_F^2. \tag{10}$$

While this constraint provides a fundamental supervision signal ensuring physical fidelity, preliminary experiments indicate that relying solely on observation consistency yields limited constraint strength. To address this and improve reconstruction quality, we further incorporate a TEP.

2) Transformation Equivariance Prior: To enhance the network's robustness and regularization, we integrate the TEP into the unsupervised training framework. Let $\mathbf{P}$ denote a randomly sampled spatial transformation operator, including rotation, flipping, and spatial shifting. In our implementation, one transformation category is uniformly sampled during training. In particular, rotation angles are uniformly selected from 90°, 180°, and 270°; flipping directions are uniformly selected from horizontal, vertical, and both directions; and spatial shifting is implemented as circular translation, with horizontal and vertical offsets uniformly sampled from 0 to 7 pixels. By applying $\mathbf{P}$ to the predicted image $\hat{\mathbf{X}}^{(T)}$ and subsequently degrading it, we obtain the transformed pseudo-observations

$$\hat{\mathbf{Y}}_2 = \hat{\mathbf{R}}\left(\hat{\mathbf{X}}^{(T)}\mathbf{P}\right), \quad \hat{\mathbf{Z}}_2 = \left(\sum_{i=1}^{C} \mathbf{M}_i \odot \left(\hat{\mathbf{X}}^{(T)}\mathbf{P}\right)_i \right)_{\downarrow 2\times 2}. \tag{11}$$

Feeding the observations $\hat{\mathbf{Y}}_2$ and $\hat{\mathbf{Z}}_2$ back into the fusion network yields a new prediction, denoted as $\hat{\mathbf{X}}_2^{(T)}$. The TEP-based loss function effectively minimizes the discrepancy between this new prediction and the transformed prediction before

$$\mathcal{L}_{TE} = \left\| \hat{\mathbf{X}}_2^{(T)} - \hat{\mathbf{X}}^{(T)}\mathbf{P} \right\|_F^2. \tag{12}$$

This loss term significantly improves the model's expressive capability and generalization by enforcing consistency across geometric transformations.

3) Total Loss: Combining both constraints, the overall unsupervised training objective is defined as

$$\mathcal{L} = \mathcal{L}_{OC} + \lambda \mathcal{L}_{TE} \tag{13}$$

where λ governs the tradeoff between the two terms. The observation consistency term ensures physical fidelity between the predicted and observed domains, while the TEP-based term enhances the regularization of the solution space. Jointly, these loss functions provide effective supervision for the optimization of the proposed UDUN.

IV. Experiments

A. Datasets

In our experiments, we employ three widely used hyperspectral datasets, i.e., CAVE [65], Pavia Center (PaviaC) [66], and Chikusei [67], respectively.

CAVE Dataset is a widely used benchmark hyperspectral dataset released by the Computer Vision Laboratory of Columbia University. It contains HSIs with a spatial resolution of 512×512 pixels and 31 spectral bands spanning wavelengths from 400 to 700 nm at 10-nm intervals. In our experiments, we select the 12th to the 27th bands to construct 16-band HSI. Each spectral channel is normalized independently by dividing its pixel values by the maximum value of that channel. The CAVE dataset consists of 32 images, among which 20 images are used for training, and the remaining 12 are used for testing.

PaviaC Dataset, collected by ROSIS-3 and released by the University of Pavia team, covers the central area of Pavia. The original data contain 102 spectral bands and have a spatial size of 1096×715 pixels with approximately 1.3-m ground sampling distance. We select the 9th–13th, 29th–33th, and 59th–64th bands to construct a 16-band HSI. The data are normalized by dividing pixel values by 2^{12}. For data splitting, the dataset is partitioned along the width dimension (splitting width = 220 pixels), resulting in a $1096 \times 495 \times 16$ subblock for training and a $1096 \times 220 \times 16$ subblock for testing.

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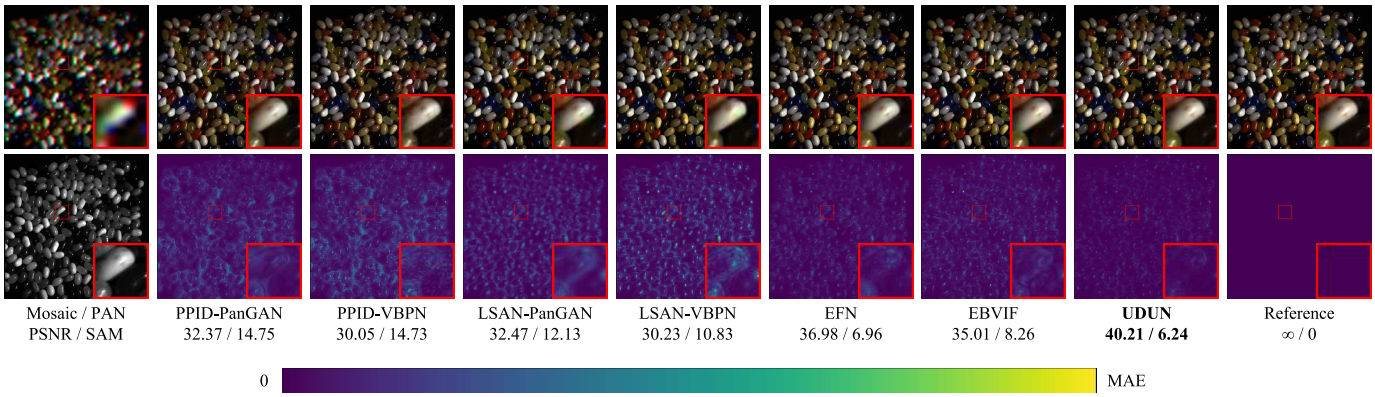


Fig. 6. Visual comparison of fusion methods on the CAVE dataset. (Top) Pseudocolor image. (Middle) MAE map. (Bottom) PSNR/SAM values computed for the displayed example. Note that the MAE maps of all the methods are scaled for better contrast.

The testing subblock is further divided into five testing images by splitting along the height dimension with a height of 220 pixels each.

Chikusei Dataset, collected by Hyperspec-VNIR-CIRIS, covers Chikusei City and contains 128 valid spectral bands with a spatial size of 2335 $\times$ 2517 pixels and a ground sampling distance of 2.5 m. Consistent with PaviaC, we select the same spectral bands to construct a 16-band HSI. The data are normalized by dividing pixel values by 2^{14}. For dataset partitioning, we split the data along the width dimension (splitting width = 400 pixels), resulting in a $2335 \times 2117 \times 16$ subblock for training and a 2335 $\times$ 400 $\times$ 16 subblock for testing. The testing subblock is further divided into six testing images by splitting along the height dimension with a height of 400 pixels each.

For all datasets, the mosaiced and PAN observations are simulated from the ground-truth HSIs using the same degradation process. In particular, an 8 $\times$ 8 SFA mosaicing pattern is first applied, followed by 2 $\times$ 2 average downsampling to generate the mosaiced images. The PAN images are obtained by applying a customized SRF, defined as $(1, 1, 2, 4, 8, 9, 10, 12, 16, 12, 10, 9, 7, 3, 2, 1)/107$, to the HSIs. For visualization, the third, eighth, and 13th bands are selected to form the three-channel pseudocolor images.

B. Training Setup

For the CAVE, PaviaC, and Chikusei datasets, both the mosaiced and PAN images are cropped into patches of size 32 $\times$ 32 and 64 $\times$ 64, respectively. The cropping stride is set to 32 $\times$ 32, 16×16, and 32 $\times$ 32 for CAVE, PaviaC, and Chikusei, respectively. The batch sizes are configured as 16, 8, and 16 for the three datasets. The network parameters are optimized using the Adam optimizer with a fixed learning rate of 10^{-4} and no learning rate decay. The total number of training iterations is set to 5.64 $\times$ 10^4, 5.78 $\times$ 10^4, and 4.40 $\times$ 10^4 for the CAVE, PaviaC, and Chikusei datasets, respectively. The fusion network is implemented in PyTorch and trained on a single NVIDIA RTX 4090 GPU. In all subsequent experiments, the step size parameter η in UDUN and the regularization weight λ are both fixed to 1.

C. Competing Methods and Evaluation Metrics

We construct comparative approaches from two perspectives, i.e., cascaded two-stage strategies and a representative end-to-end one-stage method. For the two-stage strategies, the task is decomposed into spectral demosaicing and pansharpening. In particular, the mosaiced image is first used to reconstruct an LR HSI with fully recovered spectral channels, which still maintains a 2 $\times$ 2 spatial resolution gap with respect to the PAN image. Subsequently, the LR HSI is spatially enhanced using the HR PAN image to produce the final HR HSI. For spectral demosaicing, PPID [19] and LSAN [49] are adopted. For pansharpening, PanGAN [48] and VBPN [40] are employed. In addition to the above two-stage pipelines, we further include EFN [54] and EBVIF [55] as the representative one-stage methods. For training, we train these methods using the same configurations as the proposed method, such as optimizer, learning rate, batch size, etc., to ensure fairness.

In experiments, we select four widely adopted metrics, i.e., PSNR, SSIM [68], ERGAS [69], and SAM [70], to evaluate the fusion performance.

D. Comparison Experiments

1) Qualitative Evaluation: For pseudocolor visualization, the third, eighth, and 13th bands of each reconstructed 16-band HSI are selected to form the three-channel pseudocolor images. As shown in Figs. 6–8, we present a qualitative comparison of the competing methods on the CAVE, PaviaC, and Chikusei datasets using pseudocolor images, MAE error maps, and the corresponding PSNR/SAM values computed for the displayed example and shown below each result. It can be observed that PPID-PanGAN, PPID-VBPN, LSAN-PanGAN, and LSAN-VBPN suffer from noticeable spectral and spatial distortions, indicating their limited reconstruction capability. EFN achieves improved visual quality by incorporating TEP into the training framework, which helps enhance the overall fusion performance. In comparison, the proposed method delivers the best visual results. Built upon the EFN framework, the proposed UDUN further introduces a deep unfolded network to achieve more accurate degradation kernel

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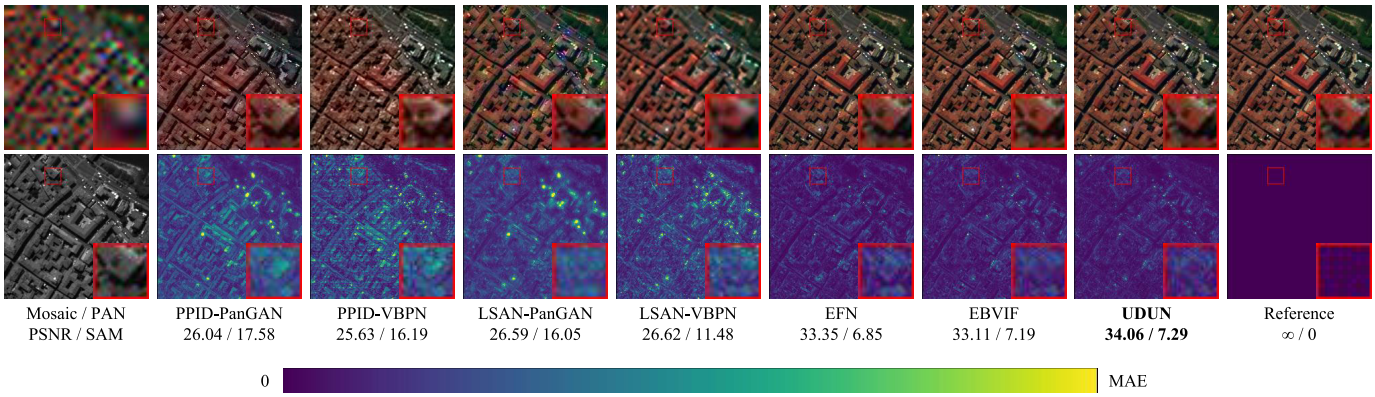


Fig. 7. Visual comparison of fusion methods on the PaviaC dataset. (Top) Pseudocolor image. (Middle) MAE map. (Bottom) PSNR/SAM values computed for the displayed example. Note that the MAE maps of all the methods are scaled for better contrast.

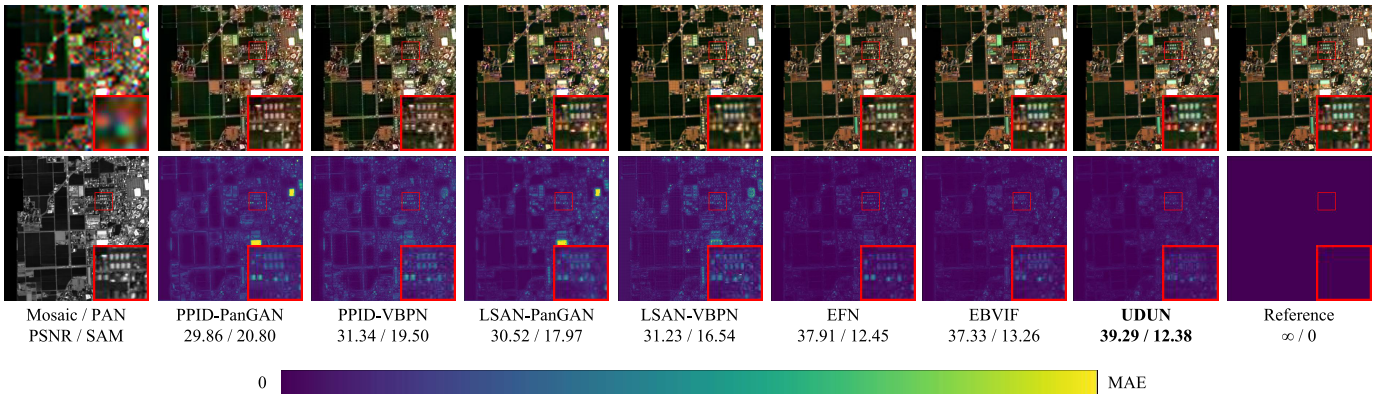


Fig. 8. Visual comparison of fusion methods on the Chikusei dataset. (Top) Pseudocolor image. (Middle). MAE map. (Bottom) PSNR/SAM values computed for the displayed example. Note that the MAE maps of all the methods are scaled for better contrast.

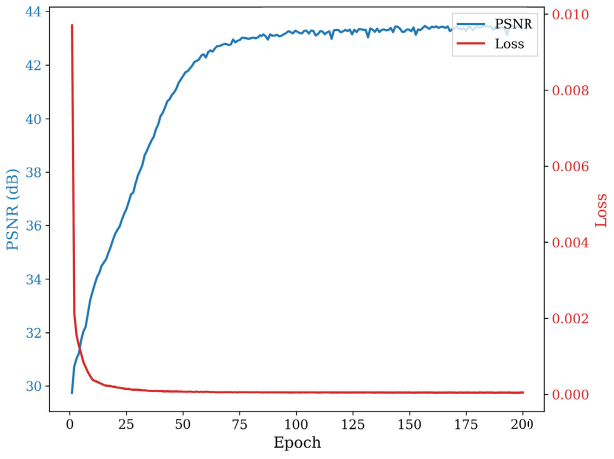


Fig. 9. Loss and PSNR curves of the proposed UDUN on the CAVE dataset during training.

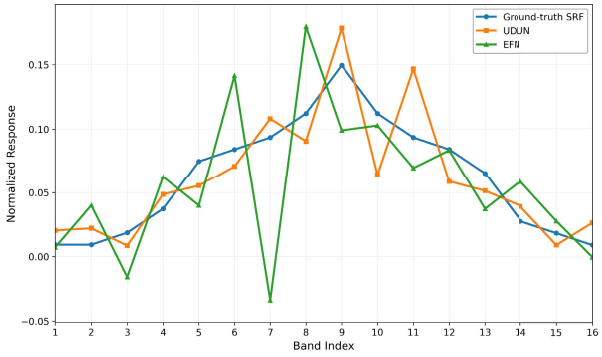


Fig. 10. Comparison of the learned SRFs of UDUN and EFN with the ground-truth SRF on the CAVE dataset. The SRF learned by UDUN is generally closer to the ground-truth SRF and exhibits more stable spectral responses across different bands.

estimation, which ultimately contributes to superior fusion performance.

In addition, we analyze the training convergence and the learned degradation function of the proposed UDUN on the CAVE dataset. As shown in Fig. 9, the total loss steadily decreases while the PSNR gradually increases and converges, indicating stable optimization behavior under the unsupervised setting. Furthermore, Fig. 10 shows the comparison of the learned SRFs of UDUN and EFN with the ground-truth SRF on the CAVE dataset. The SRF estimated by UDUN generally follows the ground-truth trend and remains more stable across spectral bands, whereas EFN shows larger fluctuations in several bands. These results indicate that the proposed degradation-aware unfolded framework can achieve stable optimization and learn a more physically consistent SRF for TEP-based regularization.

2) Quantitative Evaluation: As illustrated in Tables I–III, PPID-PanGAN, PPID-VBPN, LSAN-PanGAN, and

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TABLE I

QUANTITATIVE EVALUATION OF THE COMPETING METHODS ON THE CAVE DATASET. ↑ INDICATES HIGHER IS BETTER, WHILE ↓ INDICATES LOWER IS BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	*PSNR*↑	*SSIM*↑	*SAM*↓	*ERGAS*↓
PPID-PanGAN	34.85	0.9457	10.8828	4.0852
PPID-VBPN	33.21	0.9007	10.7382	4.7555
LSAN-PanGAN	37.10	0.9684	7.7853	3.1186
LSAN-VBPN	35.19	0.9334	9.0624	3.7457
EFN	40.55	0.9833	5.3149	2.2558
EBVIF	38.24	0.9750	6.4651	2.8130
UDUN	**43.81**	**0.9875**	**4.9873**	**1.6717**

TABLE II

QUANTITATIVE EVALUATION OF THE COMPETING METHODS ON THE PAVIAC DATASET. ↑ INDICATES HIGHER IS BETTER, WHILE ↓ INDICATES LOWER IS BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	*PSNR*↑	*SSIM*↑	*SAM*↓	*ERGAS*↓
PPID-PanGAN	26.19	0.8115	16.7863	16.7863
PPID-VBPN	26.01	0.7554	15.4840	7.4546
LSAN-PanGAN	26.60	0.8387	15.3998	6.7508
LSAN-VBPN	26.64	0.7701	10.9965	6.5875
EFN	33.39	0.9381	**6.5893**	3.2551
EBVIF	32.97	0.9381	7.0103	3.3289
UDUN	**33.94**	**0.9424**	6.9675	**3.1033**

TABLE III

QUANTITATIVE EVALUATION OF THE COMPETING METHODS ON THE CHIKUSEI DATASET. ↑ INDICATES HIGHER IS BETTER, WHILE ↓ INDICATES LOWER IS BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	*PSNR*↑	*SSIM*↑	*SAM*↓	*ERGAS*↓
PPID-PanGAN	33.25	0.8988	23.2011	8.3582
PPID-VBPN	33.37	0.8800	21.8142	8.1733
LSAN-PanGAN	33.77	0.9285	20.8042	7.7551
LSAN-VBPN	33.67	0.8476	19.4300	7.7413
EFN	40.43	0.9779	15.9417	4.0292
EBVIF	39.64	0.9752	16.4985	4.2023
UDUN	**41.42**	**0.9772**	**15.8693**	**3.5060**

LSAN-VBPN show relatively lower performance, due to significant spectral and spatial distortions. Although EFN achieves impressive results, the proposed UDUN consistently achieves superior performance in all metrics. This is attributed to the proposed approach's ability to effectively integrate spectral and spatial information, ensuring a more accurate fusion. Notably, this demonstrates the potential of the proposed approach for HR HSI fusion tasks.

E. Ablation Study of Loss Function Components

Table IV summarizes the performance of the fusion network with different combinations of the OCP and TEP loss functions

TABLE IV

PERFORMANCE COMPARISON OF THE PROPOSED FUSION NETWORK TRAINED WITH DIFFERENT PROPOSED LOSS FUNCTIONS ON THE CAVE DATASET

$\mathcal{L}_{OC}$	$\mathcal{L}_{TE}$	*PSNR*↑	*SSIM*↑	*SAM*↓	*ERGAS*↓
✓		38.4228	0.9709	6.4709	3.4897
	✓	37.8812	0.9644	7.7977	3.1946
✓	✓	**43.8110**	**0.9875**	**4.9873**	**1.6717**

TABLE V

SENSITIVITY ANALYSIS OF THE LOSS WEIGHT λ ON THE CAVE DATASET. ↑ INDICATES HIGHER IS BETTER, WHILE ↓ INDICATES LOWER IS BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

λ	*PSNR*↑	*SSIM*↑	*SAM*↓	*ERGAS*↓
0.1	43.44	0.9871	**4.8965**	1.7288
10	43.37	0.9867	5.3481	1.7317
1.0	**43.81**	**0.9875**	4.9873	**1.6717**

TABLE VI

PERFORMANCE COMPARISON OF THE PROPOSED FUSION NETWORK UNDER DIFFERENT ITERATION NUMBERS ON THE CAVE DATASET

iters	*PSNR*↑	*SSIM*↑	*SAM*↓	*ERGAS*↓
1	43.3891	0.9873	**4.8750**	1.7447
4	43.3092	0.9864	5.1529	1.7215
8	42.8064	0.9848	5.7033	1.8474
2 (UDUN)	**43.8110**	**0.9875**	4.9873	**1.6717**

on the CAVE dataset. Using only $\mathcal{L}_{OC}$, the network achieves a PSNR of 38.4228, SSIM of 0.9709, SAM of 6.4709, and ERGAS of 3.4897. With $\mathcal{L}_{TE}$ alone, the PSNR improves to 37.8812 and SAM decreases to 7.7977, highlighting the benefit of spatial fidelity via transformation equivariance. The best performance is obtained when both loss functions are used, achieving a PSNR of 43.8110, SSIM of 0.9875, SAM of 4.9873, and ERGAS of 1.6717. Significantly, while $\mathcal{L}_{OC}$ ensures adherence to the physical measurement model, it cannot uniquely determine the HR HSI due to the severe information loss in the mosaicing process. $\mathcal{L}_{TE}$ bridges this gap by imposing a strong geometric regularizer, effectively narrowing the feasible solution space to physically plausible images. This demonstrates that combining $\mathcal{L}_{OC}$ and $\mathcal{L}_{TE}$ leads to superior fusion performance by improving both physical accuracy and robustness to geometric variations.

In addition, we further analyze the sensitivity of the loss weight λ on the CAVE dataset, as summarized in Table V. Experimental results show that the performance variation under different λ settings is relatively small. Although $\lambda = 0.1$ achieves a slightly better SAM value, $\lambda = 1$ provides the best overall balance among PSNR, SSIM, and ERGAS. Therefore, we use $\lambda = 1$ in all experiments.

F. Effect of Different Iteration Numbers

Since the proposed method operates via iterative optimization, the number of iterations serves as a critical

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TABLE VII

PERFORMANCE COMPARISON OF THE PROPOSED FUSION NETWORK UNDER DIFFERENT DEGRADATION SHARING AND CONVOLUTION ARCH SETTINGS ON THE CAVE DATASET

Sharing	Structure	*PSNR*↑	*SSIM*↑	*SAM*↓	*ERGAS*↓
	1×1 Conv	41.7167	0.9810	6.0670	2.3060
	3×3 Conv	41.5605	0.9784	6.4829	2.3145
	Two 3×3 Conv	41.6255	0.9808	6.2087	2.2200
✓	1×1 Conv	43.5830	0.9868	5.2554	1.7142
✓	Two 3×3 Conv	43.2085	0.9866	5.3268	1.7421
✓	**3×3 Conv (UDUN)**	**43.8110**	**0.9875**	**4.9873**	**1.6717**

hyperparameter affecting fusion quality. To investigate this impact, we conduct an ablation study on the CAVE dataset, as detailed in Table VI. Starting with one iteration yields a solid baseline performance. However, the optimal results are observed at two iterations, where the method achieves the highest PSNR and SSIM, along with the lowest ERGAS. Conversely, increasing the iterations to four results in a slight performance dip, and a more pronounced degradation occurs at eight iterations. This decline is likely attributable to error accumulation or oversmoothing. These observations demonstrate that fusion performance does not improve monotonically with the number of iterations. Thus, selecting an appropriate stopping point is essential for maximizing reconstruction quality.

G. Investigation of Learned Degradation Function

We investigate the effect of sharing the degradation function and different convolution architectures on fusion performance, as illustrated in Table VII. Without sharing the degradation function, performance is subpar, with lower PSNR and SSIM, and higher SAM and ERGAS. This decline is primarily attributed to the discrepancy between the degradation operator used for gradient updates within the layers and the one used for loss computation, which hinders convergence. Conversely, sharing the degradation function significantly improves performance, ensuring better consistency between the network updates and loss constraints. Among the tested convolution architectures, the single 3×3 convolution strikes the best balance, yielding the highest PSNR/SSIM and lowest SAM/ERGAS. It effectively captures necessary local spatial contexts without introducing the optimization difficulties often associated with deeper stacked convolutions. These results highlight the importance of both sharing the degradation function and selecting an effective convolution architecture.

H. Computational Cost

We compare the computational cost of the proposed method with several representative approaches in terms of parameters, FLOPs, and inference time. For two-stage methods, the total cost is obtained by summing the costs of spectral demosaicing and pansharpening. All GPU-supported methods are evaluated on a single NVIDIA RTX 4090 GPU, and the results are listed in Table VIII. For the PPID-based methods, the PPID component follows its original CPU implementation. Therefore, their runtime is reported only for reference and should not be regarded as a strictly hardware-normalized comparison. As shown, different methods exhibit varying tradeoffs between complexity and efficiency. EFN achieves the lowest parameters (0.10M), FLOPs (0.10T), and the fastest inference speed (0.13 s), indicating extremely high computational efficiency. However, its fusion performance is significantly inferior to that of our method. On the contrary, the proposed method demonstrates a better balance between efficiency and performance.

TABLE VIII

COMPUTATIONAL COST COMPARISON OF THE COMPETING FUSION METHODS. NOTE THAT THE TESTED MOSAICED AND PAN IMAGE HERE HAVE A SPATIAL DIMENSION OF 512 × 512 AND 1024 × 1024, RESPECTIVELY

Method	Param (M)	FLOPs (T)	Time (s)
PPID*-PanGAN	0.19	0.43	0.85
PPID*-VBPN	2.95	0.64	0.63
LSAN-PanGAN	0.49	0.27	0.28
LSAN-VBPN	3.24	0.48	2.48
EFN	**0.10**	**0.10**	**0.13**
EBVIF	0.17	1.41	0.77
UDUN	0.20	0.32	0.15

∗ running on the CPU

V. CONCLUSION

In this article, we present UDUN for the mosaiced and PAN image fusion. To alleviate the limitations of existing methods, we identify and exploit a natural synergy between DUN and TEP. In particular, while TEP offers a powerful geometric constraint for unsupervised learning, its effectiveness is strictly contingent upon accurate degradation estimation, which is a capability inherent to DUNs but often lacking in other architectures. By integrating these two paradigms, the proposed framework utilizes the DUN to explicitly learn and refine the physical degradation process, thereby enabling the precise application of TEP. In return, TEP imposes a robust geometric prior that regularizes the solution space, effectively compensating for the weak supervision signal of the OCP in illposed inverse problems. Extensive experiments on three benchmark datasets demonstrate that the proposed method achieves superior reconstruction performance. Future work will explore extending this synergistic framework to other blind inverse problems in computational imaging.

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